Digital filters

Answers and worked solutions · Day 17 · Mon 9 Nov · pack D10

5. Practice

From your course files

T8 Q1 (tutorial).

Hint: three equations: $H(2\pi/3) = 0$, $H(0) = 1$, $b_0 = b_2$.

$h(n) = \{\underset{\uparrow}{b_0}, b_1, b_2\}$, so $H(\omega) = b_0 + b_1 e^{-j\omega} + b_2 e^{-j2\omega}$. The requirements give

$$0 = b_0 + b_1 e^{-j2\pi/3} + b_2 e^{-j4\pi/3}\ [\text{i}], \qquad 1 = b_0 + b_1 + b_2\ [\text{ii}], \qquad 0 = b_0 - b_2\ [\text{iii}].$$

With $b_0 = b_2$, [i] becomes $e^{-j2\pi/3}\left(2b_0 \cos\frac{2\pi}{3} + b_1\right) = 0$, so $b_1 = b_0$. Then [ii] gives

$$b_0 = b_1 = b_2 = \tfrac{1}{3}.$$

$$H(\omega) = \tfrac{1}{3}e^{-j\omega}\left(e^{j\omega} + 1 + e^{-j\omega}\right) = \tfrac{1}{3}(1 + 2\cos\omega)e^{-j\omega}$$

$$|H(\omega)| = \tfrac{1}{3}|1 + 2\cos\omega|, \qquad \theta(\omega) = \begin{cases} -\omega, & 1 + 2\cos\omega \geq 0 \\ -\omega \pm \pi, & \text{otherwise} \end{cases}$$

Check the sketch: $|H(0)| = 1$, a null at $\omega = 2\pi/3$, $|H(\pi)| = \frac{1}{3}$; the phase is linear with slope -1 and jumps by π at the null.

T8 Q2 (tutorial).

Hint: 5 taps, so the nulls are at $2\pi k/5$.

Geometric series: $H(z) = \dfrac{1}{5}\dfrac{1 - z^{-5}}{1 - z^{-1}}$, so

$$H(\omega) = \frac{1}{5}\frac{e^{-j5\omega/2}\left(e^{j5\omega/2} - e^{-j5\omega/2}\right)}{e^{-j\omega/2}\left(e^{j\omega/2} - e^{-j\omega/2}\right)} = \frac{1}{5}e^{-j2\omega}\frac{\sin\frac{5\omega}{2}}{\sin\frac{\omega}{2}}$$

$$|H(\omega)| = \frac{1}{5}\left|\frac{\sin(5\omega/2)}{\sin(\omega/2)}\right|, \qquad \theta(\omega) = \begin{cases} -2\omega, & \dfrac{\sin(5\omega/2)}{\sin(\omega/2)} \geq 0 \\ -2\omega \pm \pi, & \text{otherwise.} \end{cases}$$

$H(\omega) = 0$ at $\omega = 2\pi k/5$, $k = 1, 2$ on $0 \leq \omega \leq \pi$. In continuous time,

$$\Omega = \frac{2\pi k}{5} F_s = 400\pi k \text{ rad/s}, \quad \text{i.e. } F = 200k \text{ Hz}, \; k = 1, 2: \quad F = 200 \text{ Hz and } 400 \text{ Hz}.$$

T8 Q3 (tutorial).

Hint: substitute $z^{-1} = -1$ (that is, $\omega = \pi$) for the gain condition.

Null frequency: $\omega_0 = 2\pi \cdot 50/500 = \pi/5$ rad. Zeros at $e^{\pm j\omega_0}$ give

$$H(\omega) = b_0 \left(1 - 2\cos\omega_0 \, e^{-j\omega} + e^{-j2\omega}\right).$$

At $\omega = \pi$: $1 = b_0 \left(1 + 2\cos\frac{\pi}{5} + 1\right)$, so

$$b_0 = \frac{1/2}{1 + \cos\frac{\pi}{5}} = 0.2764, \qquad H(z) = \frac{1/2}{1 + \cos\frac{\pi}{5}} \left(1 - 2z^{-1}\cos\frac{\pi}{5} + z^{-2}\right) = 0.2764 \left(1 - 1.6180 z^{-1} + z^{-2}\right).$$

T8 Q4 (tutorial).

Hint: minimum phase, so the causal inverse is stable.

(a) $H(z) = 1 - 0.9z^{-4} = \dfrac{z^4 - 0.9}{z^4}$: zeros at $z_k = 0.9^{1/4} e^{j\pi k/2} = 0.9740 \, e^{j\pi k/2}$, $k = 0, 1, 2, 3$ (on the axes, just inside the unit circle), and a 4th-order pole at 0.

(b) $H(\omega) = 1 - 0.9e^{-j4\omega}$, so

$$|H(\omega)|^2 = 1 - 1.8\cos 4\omega + 0.81 = 1.81 - 1.8\cos 4\omega.$$

On $0 \le \omega \le \pi$, $e^{j\omega}$ passes close to zeros at $\omega = 0, \pi/2, \pi$ and is furthest from them at $\pi/4, 3\pi/4$. The 4th-order pole at 0 does not affect $|H|$.

ω	**0**	$\pi/4$	$\pi/2$	$3\pi/4$	π
$\lvert H(\omega)\rvert$	0.1	1.9	0.1	1.9	0.1

Sketch: a smooth periodic ripple with period $\pi/2$, minima of 0.1 (not 0, because the zeros are just inside the circle) at $0, \pi/2, \pi$, and maxima of 1.9 at $\pi/4, 3\pi/4$. It looks like two "humps" on $[0, \pi]$, a 4-tooth comb.

(c)

$$H_I(z) = H^{-1}(z) = \frac{1}{1 - 0.9z^{-4}}, \qquad \text{ROC: } |z| > 0.9^{1/4}.$$

Stable, because this ROC includes the unit circle ($H(z)$ is minimum-phase, so its causal inverse is stable).

(d) The poles of H become zeros and the zeros become poles: $H_I(z) = \dfrac{z^4}{z^4 - 0.9}$ has poles at $0.974e^{j\pi k/2}$ and a 4th-order zero at 0. $|H_I(\omega)| = 1/|H(\omega)|$:

ω	**0**	$\pi/4$	$\pi/2$	$3\pi/4$	π
$\lvert H_I(\omega)\rvert$	10	0.526	10	0.526	10

Sketch: sharp peaks of height 10 at $0, \pi/2, \pi$ (the poles are very close to the circle there), and broad valleys of 0.526 at $\pi/4, 3\pi/4$. Peaks are narrow, valleys are wide: the opposite shape to (b).

T8 Q5 (tutorial).

Hint: section 1.7. Equal degrees, so the constant is $K = 0.64/(-1.2)$.

(a) Factorise:

$$H(z) = \frac{\left(1 - 0.8z^{-1}\right)\left(1 + \frac{3}{2}z^{-1}\right)}{\left(1 - 0.8e^{j\pi/3}z^{-1}\right)\left(1 - 0.8e^{-j\pi/3}z^{-1}\right)}$$

Zeros at 0.8 and $-\frac{3}{2}$; poles at $0.8e^{\pm j\pi/3}$.

(b) Nonminimum-phase: one zero is inside and one is outside the unit circle.

(c) Because it is nonminimum-phase, its stable inverse is noncausal:

$$H_I(z) = H^{-1}(z) = \frac{1 - 0.8z^{-1} + 0.64z^{-2}}{1 + 0.7z^{-1} - 1.2z^{-2}}, \qquad \text{ROC: } 0.8 < |z| < \tfrac{3}{2}.$$

Partial fractions:

$$H_I(z) = \frac{0.3478}{1 - 0.8z^{-1}} + \frac{1.1855}{1 + \frac{3}{2}z^{-1}} - 0.5333$$

$$h_I(n) = 0.3478\,(0.8)^n u(n) - 1.1855\left(-\tfrac{3}{2}\right)^n u(-n-1) - 0.5333\,\delta(n),$$

which is absolutely summable, as expected. It is **noncausal** (two-sided).

New questions

N1.

a. Flip the sign of the odd sample: $h_{HP}(n) = \{0.25, \underset{\uparrow}{-0.5}, 0.25\}$, so $H_{HP}(z) = 0.25 - 0.5z^{-1} + 0.25z^{-2} = 0.25(1 - z^{-1})^2$. This is $H_{LP}(-z)$.

b. $H_{LP}(z) = 0.25(1 + z^{-1})^2$ has a double zero at $z = -1$. H_{HP} has a double zero at $z = +1$ (reflected through the origin).

$|H_{LP}(\omega)| = \cos^2(\omega/2)$ and $|H_{HP}(\omega)| = \sin^2(\omega/2)$.

ω	0	$\pi/2$	π
LP	1	0.5	0
HP	0	0.5	1

Both are symmetric, so both have linear phase with delay 1 sample.

N2.

a. $\omega_0 = 2\pi(50)/1000 = \pi/10 = 0.3142$ rad. $2\cos\omega_0 = 1.9021$.
$H(z) = b_0(1 - 1.9021z^{-1} + z^{-2})$. At DC ($z = 1$):
$b_0(2 - 1.9021) = b_0(0.09789) = 1$, so $b_0 = 10.216$.

$$H(z) = 10.216 - 19.432z^{-1} + 10.216z^{-2}$$

b. With $|H(\omega)| = b_0|2\cos\omega - 2\cos\omega_0|$ (from $H = b_0e^{-j\omega}(2\cos\omega - 2\cos\omega_0)$):

- 100 Hz ($\omega = \pi/5$): $|H| = 2.902$.
- 500 Hz ($\omega = \pi$): $|H| = 39.86$.

The FIR notch is so wide that, once you force DC gain 1, everything above the notch is amplified hugely. Not usable.

c. Denominator
$1 - 2(0.95)\cos(\pi/10)z^{-1} + 0.9025z^{-2} = 1 - 1.8070z^{-1} + 0.9025z^{-2}$. At $z = 1$:
$b_0\dfrac{2 - 1.9021}{1 - 1.8070 + 0.9025} = b_0\dfrac{0.09789}{0.09550} = 1$, so $b_0 = 0.9755$.

$$H(z) = 0.9755\,\frac{1 - 1.9021z^{-1} + z^{-2}}{1 - 1.8070z^{-1} + 0.9025z^{-2}}$$

d. Python: $|H| = 1.012$ at 100 Hz and 1.026 at 500 Hz. Nearly flat away from the notch. The -3 dB band is about 42.2–57.8 Hz, close to the $2(1 - r)F_s/(2\pi) = 15.9$ Hz estimate.

N3.

a. Nulls at $F_k = kF_s/(M + 1)$. Need $F_s/(M + 1) = 1$ kHz, so $M + 1 = 8$, $M = 7$. Nulls at 1, 2, 3, 4 kHz.

b. $\omega = 2\pi(500)/8000 = \pi/8$. $|H| = \dfrac{1}{8}\left|\dfrac{\sin(8\cdot\pi/16)}{\sin(\pi/16)}\right| = \dfrac{1}{8}\cdot\dfrac{1}{0.19509} = 0.6407$.
Symmetric, so linear phase with delay $M/2 = 3.5$ samples $= 3.5/8000$ s $= 0.4375$ ms.

N4.

a. $\omega_0 = 2\pi(1000)/8000 = \pi/4$. $2r\cos\omega_0 = 2(0.95)(0.7071) = 1.3435$, $r^2 = 0.9025$.

$$H(z) = \frac{b_0(1 - z^{-2})}{1 - 1.3435z^{-1} + 0.9025z^{-2}}$$

$$y(n) = 1.3435\,y(n-1) - 0.9025\,y(n-2) + b_0[x(n) - x(n-2)]$$

b. Use distances: $|1 - e^{-j2\omega_0}| = 2\sin\omega_0$, the near pole contributes $1 - r$, the far pole $|1 - re^{-j2\omega_0}| = \sqrt{1 - 2r\cos 2\omega_0 + r^2}$. At $\omega_0 = \pi/4$, $\cos 2\omega_0 = 0$:

$$|H(\omega_0)| = \frac{b_0 \cdot 2\sin(\pi/4)}{(0.05)\sqrt{1.9025}} = b_0(20.506)$$

So $b_0 = 0.04877$. $|H(0)| = |H(\pi)| = 0$ (the zeros at ± 1). Python confirms the peak is at $\omega \approx 0.787 \approx \pi/4$ with height 1.0003.

c. $H(z) = \dfrac{\sin(\pi/4)}{1 - 1.4142z^{-1} + z^{-2}}$, so
$y(n) = 1.4142\,y(n-1) - y(n-2) + 0.7071\,x(n)$. $h(n) = \sin(\pi(n+1)/4)u(n)$:
$h(0) = 0.7071$, $h(1) = 1$, $h(2) = 0.7071$, $h(3) = 0$. Period 8 samples, i.e. a 1 kHz tone at 8 kHz. Marginally stable (poles on the circle).

N5.

a. $1 - 1.5z^{-1} - z^{-2} = (1 - 2z^{-1})(1 + 0.5z^{-1})$. Zeros at 2 and -0.5. Pole at 0.4. Causal and stable (pole inside). One zero inside, one outside: **nonminimum-phase (mixed)**.

b. H_I has poles at 2 and -0.5. Causal ROC $|z| > 2$ excludes $|z| = 1$: **unstable**.

c. $H_I(z) = \dfrac{1 - 0.4z^{-1}}{(1 - 2z^{-1})(1 + 0.5z^{-1})}$, ROC $0.5 < |z| < 2$. Proper (degree 1 over degree 2), so no constant.

- A_1 (pole 2): $\dfrac{1 - 0.4/2}{1 + 0.5/2} = \dfrac{0.8}{1.25} = 0.64$
- A_2 (pole -0.5): $\dfrac{1 - 0.4/(-0.5)}{1 - 2/(-0.5)} = \dfrac{1.8}{5} = 0.36$

- Check: $0.64 + 0.36 = 1 = H_I(\infty)$.

Pole -0.5 inside the circle: causal. Pole 2 outside: anti-causal.

$$h_I(n) = 0.36(-0.5)^n u(n) - 0.64(2)^n u(-n-1)$$

Two-sided, so **non-causal**. The pole of H at 0.4 has become a zero of H_I.

d. $h_I(-2) = -0.64/4 = -0.16$, $h_I(-1) = -0.32$, $h_I(0) = 0.36$, $h_I(1) = -0.18$, $h_I(2) = 0.09$. Python: $h * h_I = \delta(n)$.

N6.

a. Pole at -0.5, zero at -2. They are reciprocals, the allpass signature (L09 slide 23 with $N = 1$, $A(z) = 1 + 0.5z^{-1}$). $H(\omega) = \dfrac{0.5 + e^{-j\omega}}{1 + 0.5e^{-j\omega}} = e^{-j\omega}\dfrac{1 + 0.5e^{j\omega}}{1 + 0.5e^{-j\omega}}$. The fraction is a number over its own conjugate, so its magnitude is 1. Hence $|H(\omega)| = 1$.

b. The zero at -2 is outside the unit circle, so it is **not** minimum phase (it is maximum phase). The causal inverse has a pole at -2: unstable. Its stable inverse is anti-causal.

c. Let $\phi(\omega) = \angle(1 + ae^{-j\omega}) = -\tan^{-1}\dfrac{a\sin\omega}{1 + a\cos\omega}$. Then $\theta(\omega) = -\omega - 2\phi(\omega)$ (numerator angle is $-\phi$). Differentiating, $\dfrac{d\phi}{d\omega} = -\dfrac{a\cos\omega + a^2}{1 + 2a\cos\omega + a^2}$, so

$$\tau = 1 + 2\frac{d\phi}{d\omega} = \frac{1 + 2a\cos\omega + a^2 - 2a\cos\omega - 2a^2}{1 + 2a\cos\omega + a^2} = \frac{1 - a^2}{1 + 2a\cos\omega + a^2}.$$

With $a = 0.5$: $\tau(0) = 0.75/2.25 = 0.333$, $\tau(\pi/2) = 0.75/1.25 = 0.6$, $\tau(\pi) = 0.75/0.25 = 3$ samples (matches `scipy.signal.group_delay`). The group delay varies with ω, which is exactly why an allpass works as a phase equaliser: it changes delay without changing magnitude.

6. Quick recall check

Question

1. Yes. It is symmetric ($h(n) = h(4-n)$). Group delay $= (5-1)/2 = 2$ samples.
2. $F_s/(M+1) = 2000/10 = 200$ Hz.
3. $H_{HP}(z) = \dfrac{1 - 2z^{-1} + z^{-2}}{1 + 0.5z^{-1} + 0.2z^{-2}}$ (odd powers flip sign).
4. Non-causal (two-sided), ROC $0.5 < |z| < 1.2$. H is nonminimum-phase.
5. On the unit circle at $e^{\pm j\omega_0}$. $h(n)$ is a sinusoid that never decays, so it is not absolutely summable.